

MongoDB Integration Guide

1. Install dependencies

```
pip install pymongo python-dotenv
```

2. Add to .env

```
MONGO_URI=mongodb+srv://<user>:<password>@cluster0.mongodb.net
MONGO_DB_NAME=visioninspect_ai
```

Never commit the real .env — only .env.example with placeholder values.

3. Import in your route file

```
from services.inference_log_service import save_inference_result, get_inference_result
```

4. Call it after your existing SQL insert

Right after you commit the inspection row and have its ID:

```
inspection_id = ... # from your SQL insert

save_inference_result(
    inspection_id=inspection_id,
    model_version="patchcore-demo",
    anomaly_score=score,
    prediction=status,
    heatmap_path="uploads/demo_heatmap.png",
    mask_path="uploads/demo_mask.png",
    defects_detected=[],
    processing_time_ms=300
)
```

5. Retrieve later

```
doc = get_inference_result(inspection_id)
# returns the Mongo document, or None if not found
```

Suggested route:

```
@inspection_bp.route("/<int:inspection_id>/raw", methods=["GET"])
def get_raw(inspection_id):
    return get_inference_result(inspection_id)
```

6. What this does NOT touch

- No changes to your users table, auth, JWT, or existing SQL schema.
- No changes to any of your 12 existing REST endpoints.
- Purely additive — one new collection, one new optional route.

7. Files you're receiving

```
database/mongo_connection.py    -> Atlas connection handle
services/inference_log_service.py -> save_inference_result() / get_inference_result()
```

Drop both files into your existing folder structure as-is. Merge routes/inspection.py changes yourself when convenient — shown above, small diff.